AI-Based Smart Greenhouse Automation System

A PROJECT REPORT

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in partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

KNOWLEDGE INSTITUTE OF TECHNOLOGY,SALEM

ANNA UNIVERSITY::CHENNAI 600 025

JUN 2026

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ACKNOWLEDGEMENT

We thank our beloved Executive Chairman and President **Dr.PSS.SRINIVASAN, B.E., M.Tech., Ph.D.,** and all the members of Knowledge Institute of Technology Trust at this high time for providing us with excess of facilities to complete the project successfully.

Abstract

The AI-Based Smart Greenhouse Automation System is an intelligent Internet of Things (IoT) solution designed to automate greenhouse monitoring and irrigation for improved crop growth and efficient resource utilization. The system is built around an ESP32 microcontroller and incorporates a DHT22 sensor to measure temperature and humidity, a soil moisture sensor to monitor soil water content, a BH1750 sensor to measure light intensity, and a water level sensor to monitor the water availability in the storage tank. The ESP32 continuously collects environmental data from these sensors and automatically controls the irrigation system by operating a water pump through a relay module whenever soil moisture falls below the required level. The sensor readings and system status are transmitted to the Blynk IoT cloud platform, enabling users to monitor greenhouse conditions remotely in real time through a smartphone or web dashboard. To enhance the intelligence of the system, historical sensor data collected from the greenhouse is used to train an Artificial Intelligence (AI) model. The trained model analyzes environmental conditions and predicts optimal irrigation decisions based on learned patterns, making the system more adaptive and efficient than conventional rule-based automation. This approach helps reduce water wastage, minimizes manual intervention, and supports healthy plant growth. The proposed system is cost-effective, scalable, and suitable for home gardens, educational projects, research greenhouses, and small-scale agricultural applications. By integrating IoT, cloud connectivity, and Artificial Intelligence, the project demonstrates a practical smart agriculture solution that promotes sustainable farming practices and efficient water management.

KEYWORDS:

Artificial Intelligence (AI) ,Internet of Things (IoT) ,Smart Greenhouse , ESP32 Microcontroller ,Automatic Irrigation ,Soil Moisture Sensor ,DHT22 Sensor ,BH1750 Light Sensor, Water Level Sensor ,Blynk IoT

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CHAPTER 1

INTRODUCTION

1.1 Introduction to Smart Greenhouse Automation

Agriculture plays an important role in food production and economic development. The growth of crops depends on environmental conditions such as temperature, humidity, soil moisture, and light intensity. In greenhouse cultivation, these conditions need to be monitored and maintained properly for healthy plant growth.

Traditional greenhouse systems mainly depend on manual monitoring and irrigation. This requires continuous human effort and may result in improper water usage. The integration of Internet of Things (IoT) and Artificial Intelligence (AI) provides an effective solution for automated greenhouse management.

The proposed AI-Based Smart Greenhouse Automation System uses an ESP32 microcontroller with sensors to monitor temperature, humidity, soil moisture, light intensity, and water level. The system automatically controls irrigation using a water pump and relay module. Sensor data is also transmitted to the Blynk IoT platform for remote monitoring.

1.2 Need for Smart Greenhouse Automation

Smart greenhouse automation is required to improve crop management and efficient utilization of resources. Manual monitoring of greenhouse conditions requires time and continuous human involvement.

- To continuously monitor greenhouse conditions.
- To reduce manual intervention.
- To provide automatic irrigation based on soil moisture.

- To reduce unnecessary water consumption.
- To monitor water availability in the storage tank.
- To provide remote monitoring through IoT.
- To utilize sensor data for intelligent irrigation decisions.

1.3 Problems with Conventional Greenhouse Monitoring

Conventional greenhouse monitoring systems have several limitations. Manual observation requires continuous human involvement and may not provide accurate or timely control of irrigation.

- Manual monitoring of temperature, humidity, and soil moisture.
- Improper irrigation due to lack of soil moisture monitoring.
- Water wastage caused by unnecessary irrigation.
- Difficulty in monitoring the greenhouse remotely.
- Dependence on human intervention for pump operation.
- Lack of historical data-based decision-making.
- Insufficient monitoring of water storage level.

These limitations create the need for an automated system capable of monitoring greenhouse conditions and controlling irrigation efficiently.

1.4 Importance of IoT and Artificial Intelligence in Agriculture

IoT enables sensors, controllers, and cloud platforms to communicate and exchange information. In the proposed system, the ESP32 collects sensor data and transmits it to the Blynk IoT platform, allowing users to monitor greenhouse conditions remotely.

AI improves the system by analyzing historical sensor data and identifying patterns related to environmental conditions and irrigation requirements. The

trained model can support intelligent irrigation decisions instead of relying only on fixed conditions.

The integration of IoT and AI provides the following benefits:

- Real-time monitoring.
- Remote access to greenhouse data.
- Automatic irrigation control.
- Reduced manual intervention.
- Efficient water management.
- Data-based intelligent decision-making.
- Improved greenhouse management.

Thus, IoT provides connectivity and monitoring, while AI provides data analysis and intelligent decision-making, making the greenhouse system more efficient and adaptable.

CHAPTER 2

LITERATURE SURVEY

The increasing demand for efficient agricultural practices and sustainable resource utilization has encouraged researchers to develop smart greenhouse and precision agriculture technologies. IoT-based greenhouse systems have gained attention because they enable real-time monitoring of environmental conditions and automatic control of agricultural operations. The integration of sensors, microcontrollers, cloud platforms, and automation techniques provides an effective approach for improving greenhouse management.

Recent developments have also focused on combining Artificial Intelligence with IoT systems. AI-based approaches can analyze historical environmental data and identify patterns related to crop requirements. Such systems can improve irrigation decisions and reduce unnecessary resource consumption. The following sections discuss important technologies related to the proposed smart greenhouse automation system.

2.1 IoT-Based Greenhouse Monitoring

IoT-based greenhouse monitoring systems use sensors to continuously measure environmental parameters such as temperature, humidity, soil moisture, and light intensity. A microcontroller collects and processes the sensor data and can transmit the information to a cloud platform for remote monitoring.

These systems reduce the requirement for manual observation and provide real time information about greenhouse conditions. IoT technology also allows users to monitor agricultural environments through smartphones or web-based dashboards.

2.2 Automatic Irrigation Systems

Automatic irrigation systems are designed to provide water according to the requirements of crops. Soil moisture sensors are commonly used to determine the moisture condition of the soil. When the moisture level decreases below the required condition, a controller can activate a water pump through a relay module.

Such systems help reduce unnecessary irrigation and improve water utilization. Automatic irrigation is particularly useful in greenhouse environments where maintaining suitable soil moisture is important for healthy plant growth.

2.3 Environmental Monitoring Sensors

Different sensors are used to monitor the environmental conditions required for greenhouse cultivation. Temperature and humidity sensors provide information about the surrounding atmospheric conditions, while soil moisture sensors determine the water content of the soil. Light sensors measure the intensity of available light, and water level sensors monitor the availability of water in the storage tank.

The combination of these sensors provides a more complete understanding of the greenhouse environment and supports automatic system control.

2.4 Artificial Intelligence in Smart Agriculture

Artificial Intelligence has been increasingly applied in agriculture for data analysis and decision-making. Historical data obtained from sensors can be analyzed to identify relationships between environmental conditions and crop requirements.

In greenhouse automation, AI can support irrigation decisions by analyzing parameters such as soil moisture, temperature, humidity, and light intensity. This

approach can make the system more adaptive compared with conventional systems that depend only on fixed threshold conditions.

2.5 Research Gap and Motivation

Existing greenhouse automation systems mainly focus on sensor-based monitoring and rule-based automatic irrigation. Although these systems can reduce manual intervention, their decisions are generally based on predefined threshold values and may not utilize historical environmental data effectively.

The proposed project aims to address this limitation by integrating IoT-based monitoring with Artificial Intelligence**.** The system uses ESP32, environmental sensors, automatic irrigation, and the Blynk IoT platform, while historical sensor data is used for AI-based analysis. This combination provides a foundation for developing a more intelligent and adaptive greenhouse automation system.

CHAPTER 3

PRINCIPLES OF AI-BASED SMART GREENHOUSE AUTOMATION

3.1 Introduction

The AI-Based Smart Greenhouse Automation System works based on the principles of sensor-based monitoring, automatic control, IoT communication, and Artificial Intelligence. Sensors continuously measure the environmental conditions inside the greenhouse, and the ESP32 processes the collected information.

Based on the soil moisture condition, the irrigation system is automatically controlled using a relay and water pump. The sensor data can also be transmitted to the Blynk IoT platform for remote monitoring. Historical sensor data is used for AI-based analysis to support intelligent irrigation decisions.

3.2 Environmental Parameter Monitoring

Environmental conditions have a direct influence on plant growth. The proposed system monitors important parameters such as temperature, humidity, soil moisture, light intensity, and water level.

The DHT22 sensor measures temperature and humidity, the soil moisture sensor measures the moisture content of the soil, the BH1750 sensor measures light intensity, and the water level sensor monitors the water available in the storage tank.

The collected data is sent to the ESP32 for processing and monitoring.

3.3 Automatic Irrigation Principle

Automatic irrigation is based mainly on the moisture condition of the soil. When the soil moisture falls below the required level, the ESP32 sends a control signal to the relay module.

The relay activates the water pump, which supplies water to the plants. When the required soil moisture condition is reached, the pump can be switched off. This method helps avoid unnecessary irrigation and improves water utilization.

3.4 IoT-Based Monitoring Principle

The ESP32 provides wireless connectivity between the greenhouse system and the Blynk IoT platform. The sensor readings collected by the ESP32 are transmitted to the cloud platform through the internet.

The user can view parameters such as temperature, humidity, soil moisture, light intensity, and system status through a smartphone or web dashboard. This enables real-time remote monitoring of the greenhouse.

3.5 Artificial Intelligence-Based Decision Making

Artificial Intelligence is used to analyze historical sensor data collected from the greenhouse. The data contains environmental parameters and corresponding irrigation conditions.

The AI model is trained using this historical data to identify patterns between environmental conditions and irrigation requirements. After training, the model can support irrigation decisions based on the learned patterns.

The AI-based approach provides an additional level of intelligence compared with conventional systems that depend only on fixed threshold values.

3.6 Importance of Smart Greenhouse Automation

The integration of sensors, ESP32, automatic irrigation, IoT, and AI provides an intelligent approach to greenhouse management. The system can continuously monitor environmental conditions, reduce manual intervention, and improve water management.

The proposed system also provides a foundation for developing scalable smart agriculture applications by integrating additional sensors and intelligent control functions in the future.

CHAPTER 4

SYSTEM DESIGN AND COMPONENT DESCRIPTION

4.1 Introduction

The AI-Based Smart Greenhouse Automation System is designed to monitor important environmental conditions and provide automatic irrigation for maintaining a suitable environment for plant growth. The system continuously measures temperature, humidity, soil moisture, light intensity, and water availability using different sensors. The collected data is processed through an ESP32 microcontroller.

The sensor readings are transmitted to the Blynk cloud platform, enabling remote monitoring and data visualization through a mobile application or web dashboard. The system also incorporates an automatic irrigation mechanism in which the ESP32 controls a water pump through a relay module whenever the soil moisture falls below the required level.

To improve the intelligence of the system, historical sensor data is collected and analyzed using an Artificial Intelligence model. The AI model identifies patterns in environmental conditions and supports suitable irrigation decisions. This chapter describes the overall system design and the hardware components used to achieve efficient monitoring, automatic irrigation, IoT connectivity, and AI-based decision-making.

4.2 Overall System Architecture

The overall architecture of the AI-Based Smart Greenhouse Automation System consists of sensing, processing, communication, display, irrigation, and AI analysis units working together to monitor and control greenhouse conditions. The DHT22 sensor measures temperature and humidity, the soil moisture sensor monitors the moisture content of the soil, the BH1750 sensor measures light

intensity, and the water level sensor monitors the availability of water in the storage tank.

The ESP32 acts as the central processing unit and receives the data from all the sensors. The collected information is processed to determine the current greenhouse conditions. When the soil moisture falls below the required level, the ESP32 activates the relay module, which switches ON the water pump and supplies water to the plants.

The ESP32 also transmits the sensor data to the Blynk cloud platform through its built-in Wi-Fi capability. Users can remotely monitor greenhouse conditions through a mobile application or web dashboard. Historical sensor data is used to train an AI model, which analyzes environmental patterns and supports intelligent irrigation decisions. This architecture provides continuous monitoring, automatic irrigation, remote accessibility, and intelligent control for smart greenhouse applications.

4.3 Block Diagram Description

The block diagram of the AI-Based Smart Greenhouse Automation System consists of an ESP32 microcontroller, DHT22 sensor, soil moisture sensor, BH1750 light sensor, water level sensor, relay module, water pump, Blynk cloud platform, and AI analysis unit.

The DHT22 sensor continuously measures temperature and humidity, while the soil moisture sensor measures the moisture condition of the soil. The BH1750 sensor measures light intensity and the water level sensor monitors the availability of water in the storage tank. All the sensor readings are sent to the ESP32, which acts as the central processing unit.

The ESP32 processes the sensor readings and controls the irrigation system according to the soil moisture condition. When the soil moisture falls below the

required level, the ESP32 activates the relay module, which switches ON the water pump. The pump supplies water to the plants through the irrigation pipe.

The ESP32 also transmits the sensor data to the Blynk cloud platform through Wi-Fi, enabling remote monitoring. Historical sensor data is collected and provided to the AI model for analysis. The AI model identifies patterns in the environmental data and supports suitable irrigation decisions. This block diagram illustrates the complete flow of sensing, processing, communication, and automatic control within the system.

4.4 Components List

S. No.	Component Name	Quantity
1	ESP32 DevKit V1	1
2	DHT22 Temperature and Humidity Sensor	1
3	Capacitive Soil Moisture Sensor	1
4	BH1750 Light Sensor	1
5	Water Level Sensor	1
6	4-Channel Relay Module	1
7	5V DC Water Pump	1
8	5V DC Cooling Fan	1
9	5V LED Grow Light	1
10	5V 3A Power Adapter	1
11	Water Storage Tank	1
12	Irrigation Pipe	1 set
13	Breadboard and Jumper Wires	1 set

4.5 Component Description

4.5.1 ESP32 DevKit V1

The ESP32 DevKit V1 is the main microcontroller used in the proposed system. It is selected because of its built-in Wi-Fi connectivity, multiple GPIO pins, processing capability, and suitability for IoT applications. The ESP32 receives data from the connected sensors and processes the environmental information.

The microcontroller also controls the relay module according to the programmed conditions and transmits the collected data to the Blynk cloud platform. Its wireless communication capability makes it suitable for remote greenhouse monitoring and automation.



4.5.2 DHT22 Temperature and Humidity Sensor

The DHT22 sensor is used to measure temperature and relative humidity inside the greenhouse. Temperature and humidity are important environmental parameters that influence plant growth and development.

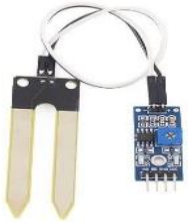
The sensor provides digital output to the ESP32, allowing continuous monitoring of greenhouse conditions. The measured values are transmitted to the Blynk platform and can also be used as input data for further analysis.



4.5.3 Capacitive Soil Moisture Sensor

The capacitive soil moisture sensor is used to measure the moisture content of the soil. Soil moisture is an important parameter for determining the irrigation requirement of plants.

The sensor provides moisture information to the ESP32. When the soil moisture falls below the required condition, the ESP32 activates the relay module and operates the water pump. This helps provide water according to the actual soil condition and reduces unnecessary water consumption.



4.5.4 BH1750 Light Sensor

The BH1750 is a digital light intensity sensor used to measure the amount of light available inside the greenhouse. Light is an essential factor for plant growth and photosynthesis.

The sensor provides light intensity readings to the ESP32. The collected data can be monitored through the IoT platform and used along with other environmental parameters for greenhouse analysis and AI-based decision-making.



4.5.5 Water Level Sensor

The water level sensor is used to monitor the amount of water available in the storage tank. It helps determine whether sufficient water is available for the irrigation process.

The sensor provides information about the water level to the ESP32. Monitoring water availability helps improve the reliability of the irrigation system and prevents unnecessary operation of the pump when water is insufficient.



4.5.6 Relay Module

The relay module is used as an electrical switching device to control the water pump using the ESP32. Since the ESP32 cannot directly provide the required power to operate the pump, the relay provides the required switching interface.

When irrigation is required, the ESP32 sends a control signal to the relay module. The relay then switches ON the water pump. The pump is switched OFF when irrigation is no longer required.



4.5.7 Water Pump

The DC water pump is used to transfer water from the storage tank to the plants. It is controlled automatically through the relay module according to the soil moisture condition.

When the soil moisture falls below the required level, the ESP32 activates the relay and the pump supplies water through the irrigation pipe. This provides automatic irrigation and reduces manual intervention.



4.5.8 Blynk IoT Platform

The Blynk IoT platform is used for remote monitoring of the greenhouse system. The ESP32 transmits the sensor readings to the Blynk cloud platform through Wi-Fi.

The user can view parameters such as temperature, humidity, soil moisture, light intensity, and water level through a mobile application or web dashboard. This enables real-time monitoring of greenhouse conditions from a remote location.



4.5.9 Power Supply Unit

The power supply unit provides the required electrical power to the ESP32, sensors, relay module, and other connected components. A stable power supply is essential for reliable sensor readings and continuous system operation.The system can be powered using a suitable DC power adapter. Proper power distribution and regulation are considered to ensure safe operation of the electronic components.



4.6 Design Considerations

Several factors were considered during the design of the AI-Based Smart Greenhouse Automation System to ensure reliable performance, affordability, and ease of use. Low-cost and readily available components such as the ESP32, DHT22 sensor, soil moisture sensor, BH1750 sensor, water level sensor, relay module, and water pump were selected for the system.

Another important consideration was the integration of IoT technology for remote monitoring. The ESP32's built-in Wi-Fi capability enables communication with the Blynk cloud platform, allowing users to access real-time greenhouse information remotely. The system was also designed with automatic irrigation to respond to changes in soil moisture conditions.

AI-based analysis was considered to improve the intelligence of the system by utilizing historical sensor data. Reliability, scalability, water efficiency, energy consumption, ease of maintenance, and component compatibility were considered throughout the design process to ensure that the system can effectively support smart greenhouse and agricultural applications.

4.7 System Advantages

- Provides real-time monitoring of temperature and humidity.
- Monitors soil moisture continuously for irrigation control.
- Measures light intensity inside the greenhouse.
- Monitors water availability in the storage tank.
- Automatically operates the water pump based on soil moisture conditions.
- Enables remote monitoring through the Blynk IoT platform.
- Uses low-cost and readily available components.
- Can be expanded for different greenhouse and agricultural applications.

CHAPTER 5

CIRCUIT DESIGN AND CONNECTIONS

5.1 Introduction

The circuit design of the AI-Based Smart Greenhouse Automation System is developed to provide proper communication between the sensors, ESP32 microcontroller, relay module, water pump, and other output devices. The circuit enables the ESP32 to collect environmental data and control the greenhouse operations according to the programmed conditions.

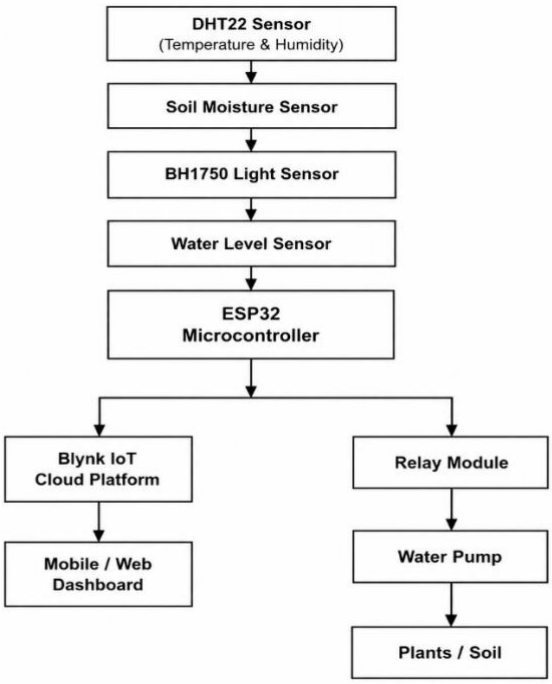
The DHT22 sensor, soil moisture sensor, BH1750 light sensor, and water level sensor are connected to the ESP32 for environmental monitoring. The relay module is connected to the ESP32 to control the water pump. The ESP32 also communicates with the Blynk IoT platform through its built-in Wi-Fi capability. Proper circuit connections are essential to ensure accurate sensor readings, reliable control, and safe operation of the complete system.

5.2 Circuit Diagram

The circuit diagram represents the electrical connections between the ESP32 microcontroller, sensors, and output devices used in the system. The DHT22 sensor is connected to the ESP32 to measure temperature and humidity, while the soil moisture sensor monitors the moisture condition of the soil. The BH1750 sensor is connected through the I²C communication interface to measure light intensity, and the water level sensor monitors the availability of water in the storage tank.

The relay module is connected to the ESP32 to control the water pump according to the irrigation requirement. When irrigation is required, the relay activates the pump to supply water to the plants. The complete circuit is powered using a suitable DC power supply, and all components are connected according to their

operating voltage and electrical requirements to ensure reliable and safe system operation.



5.3 Electrical Connections

The electrical connections are made using suitable connecting wires to ensure reliable communication between the ESP32 microcontroller, sensors, relay module, and other components. The DHT22 sensor, soil moisture sensor, BH1750 light sensor, and water level sensor are connected to the appropriate input pins of the ESP32 for collecting environmental data.

The output of the ESP32 is connected to the relay module, which controls the water pump according to the irrigation requirement. The BH1750 sensor communicates with the ESP32 through the I^2C interface, while the other sensors are connected according to their respective signal requirements. Proper polarity, common grounding, and suitable power connections are maintained throughout the circuit to ensure safe and reliable operation.

5.4 Sensor Wiring and Connections

The DHT22 sensor is connected to the ESP32 to measure temperature and humidity inside the greenhouse. Its power, ground, and data terminals are connected to the corresponding ESP32 connections. The soil moisture sensor is connected to an analog input of the ESP32 to measure the moisture condition of the soil.

The BH1750 light sensor is connected to the ESP32 through the I²C communication interface using the SDA and SCL lines. The water level sensor is connected to an appropriate input of the ESP32 to monitor the availability of water in the storage tank. Proper wiring and secure connections are maintained to obtain accurate sensor readings and reliable data transmission.

5.5 Relay Module and Water Pump Connections

The relay module is connected to the ESP32 to control the operation of the water pump. The control input of the relay is connected to a digital output pin of the ESP32, while the pump is connected to the switching terminals of the relay. The relay provides electrical isolation and allows the low-power ESP32 control signal to operate the water pump safely.

When the soil moisture falls below the required level, the ESP32 sends a control signal to the relay module. The relay switches ON the water pump and water is supplied to the plants through the irrigation pipe. When the required soil moisture condition is reached, the relay switches OFF the pump. This arrangement provides automatic and efficient irrigation based on the soil moisture condition.

CHAPTER 6

WORKING PRINCIPLE

6.1 Introduction

The working principle of the AI-Based Smart Greenhouse Automation System is based on continuous environmental monitoring, automatic irrigation, IoT communication, and AI-based decision-making. The system uses sensors to collect information about temperature, humidity, soil moisture, light intensity, and water availability.

The ESP32 microcontroller processes the collected sensor data and controls the irrigation system according to the soil moisture condition. The sensor readings are also transmitted to the Blynk cloud platform for remote monitoring. Historical sensor data is analyzed using an AI model to support intelligent irrigation decisions and improve the efficiency of the system.

6.2 Environmental Monitoring Process

The environmental monitoring process begins with the collection of data from the sensors installed inside the greenhouse. The DHT22 sensor measures temperature and humidity, while the soil moisture sensor measures the moisture content of the soil. The BH1750 sensor measures light intensity, and the water level sensor monitors the availability of water in the storage tank.

The collected sensor data is sent to the ESP32 microcontroller for processing. The ESP32 continuously monitors the environmental conditions and uses the obtained information for automatic system control and data transmission.

6.3 Automatic Irrigation Process

The automatic irrigation process is mainly controlled based on the soil moisture condition. When the soil moisture falls below the required level, the ESP32 identifies that irrigation is required and sends a control signal to the relay module.

The relay module switches ON the water pump, which supplies water from the storage tank to the plants through the irrigation pipe. When the required soil moisture condition is reached, the ESP32 switches OFF the relay and stops the water pump. This process helps reduce unnecessary water consumption and manual intervention.

6.4 IoT and AI-Based Monitoring Process

The ESP32 transmits the collected sensor data to the Blynk cloud platform through Wi-Fi. The user can monitor temperature, humidity, soil moisture, light intensity, water level, and system status through a mobile application or web dashboard.

The historical sensor data collected during system operation is used for AI-based analysis. The AI model analyzes the environmental parameters and identifies patterns related to irrigation requirements. The trained model supports suitable irrigation decisions based on the learned environmental conditions.

CHAPTER 7

IMPLEMENTATION AND TESTING

7.1 Introduction

The AI-Based Smart Greenhouse Automation System is implemented using an ESP32 microcontroller, environmental sensors, relay module, water pump, and IoT cloud platform. The system is developed to continuously monitor greenhouse conditions and automatically control irrigation based on soil moisture.

During implementation, the DHT22, soil moisture sensor, BH1750, and water level sensor are connected to the ESP32. The collected readings are processed by the controller and transmitted to the Blynk IoT platform. The irrigation system is tested by operating the water pump through the relay module according to the soil moisture condition.

7.2 Hardware Implementation

The hardware implementation consists of connecting all sensors and output devices to the ESP32 microcontroller. The DHT22 sensor measures temperature and humidity, while the soil moisture sensor measures the moisture content of the soil. The BH1750 measures light intensity, and the water level sensor monitors the water available in the storage tank.

The relay module is connected to the ESP32 to control the water pump. When irrigation is required, the relay activates the pump and water is supplied to the plants. The components are assembled and tested to ensure proper communication and reliable operation.

7.3 IoT Implementation

The ESP32 uses its built-in Wi-Fi capability to connect the greenhouse system to the Blynk IoT cloud platform. The sensor readings are transmitted to the Blynk dashboard, where users can monitor the greenhouse conditions remotely.

The dashboard provides information such as temperature, humidity, soil moisture, light intensity, water level, and irrigation status. This allows the user to observe the greenhouse conditions without being physically present at the location.

7.4 AI Model Implementation

Historical sensor data collected from the greenhouse is used for AI model development. The data includes environmental parameters such as temperature, humidity, soil moisture, and light intensity along with the corresponding irrigation conditions.

The collected data is processed and used to train the AI model. After training, the model analyzes the environmental conditions and supports irrigation decisions. The AI implementation provides an intelligent approach to irrigation compared with a system based only on fixed threshold values.

7.5 System Testing

The complete system is tested by checking each sensor and output device individually and then testing the complete system together. The DHT22 is tested for temperature and humidity measurement, while the soil moisture sensor is tested under different soil moisture conditions.

The BH1750 and water level sensor are tested to verify their respective measurements. The relay and water pump are tested by changing the soil moisture condition. The Blynk connection is also tested to ensure that the sensor readings are correctly displayed on the dashboard.

7.6 Overall System Operation

After successful testing, all the components are operated together as a complete smart greenhouse system. The sensors continuously collect environmental information and send it to the ESP32. The ESP32 processes the readings and controls the irrigation system according to the required conditions.

At the same time, the sensor data is transmitted to the Blynk platform for remote monitoring. Historical data is used for AI-based analysis to support intelligent irrigation decisions. The combined operation of sensing, processing, IoT monitoring, AI analysis, and automatic irrigation provides an integrated greenhouse automation system.

CHAPTER 8

COST ESTIMATION

8.1 Introduction

The cost estimation of the AI-Based Smart Greenhouse Automation System includes the expenses of the electronic components, sensors, irrigation components, power supply, and other materials required for developing the prototype. The system is designed using low-cost and easily available components to make it suitable for small-scale greenhouse and agricultural applications.

The major cost is associated with the ESP32 microcontroller, environmental sensors, relay module, water pump, power supply, and supporting components.

8.2 Cost Estimation Methodology

The cost estimation methodology involves identifying all the hardware and supporting components required for the development of the AI-Based Smart Greenhouse Automation System. The cost of each component is estimated based on its required quantity and approximate market price. The total project cost is calculated by adding the individual costs of all components.

The major cost categories include the ESP32 microcontroller, environmental sensors, relay module, water pump, cooling fan, LED grow light, power supply, water storage tank, irrigation pipe, and connecting accessories. The estimated cost helps in evaluating the overall affordability of the proposed system and ensures that the prototype can be developed using low-cost and easily available components.

8.3 Component Cost Estimation

S. No.	Component Name	Quantity	Approx. Unit Cost (₹)	Total Cost (₹)
1	ESP32 DevKit V1	1	400	400
2	DHT22 Temperature and Humidity Sensor	1	300	300
3	Capacitive Soil Moisture Sensor	1	150	150
4	BH1750 Light Sensor	1	120	120
5	Water Level Sensor	1	80	80
6	4-Channel Relay Module	1	150	150
7	5V DC Water Pump	1	150	150
8	5V DC Cooling Fan	1	120	120
9	5V LED Grow Light	1	150	150
10	5V 3A Power Adapter	1	300	300
11	Water Storage Tank	1	250	250
12	Irrigation Pipe	1 Set	100	100
13	Breadboard and Jumper Wires	1 Set	200	200
	Total Estimated Cost			₹2,470

CHAPTER 9

RESULTS AND ANALYSIS

9.1 Introduction

The AI-Based Smart Greenhouse Automation System was implemented and tested to evaluate its performance in environmental monitoring, automatic irrigation, IoT-based monitoring, and AI-based decision making. The system successfully integrates the ESP32 microcontroller with different sensors, a relay-controlled water pump, and the Blynk IoT platform. The results obtained from the prototype are analyzed based on the response of the sensors and the operation of the irrigation system**.**

9.2 Sensor Monitoring Results

The DHT22 sensor continuously measures temperature and humidity inside the greenhouse. The soil moisture sensor provides information about the water content of the soil, while the BH1750 sensor measures the light intensity. The water level sensor monitors the availability of water in the storage tank. These sensor values are collected by the ESP32 and used for monitoring and control.

The sensor readings help in understanding the environmental conditions of the greenhouse and provide the required information for automatic irrigation.

9.3 Automatic Irrigation Results

The automatic irrigation system operates according to the soil moisture condition. When the soil moisture level falls below the required level, the ESP32 activates the relay module, which switches ON the water pump. Water is supplied to the plants through the irrigation pipe. When sufficient moisture is detected, the pump is switched OFF.

This operation reduces unnecessary watering and minimizes the need for manual intervention. The water level sensor also helps monitor the availability of water in the storage tank.

9.4 IoT Monitoring Results

The ESP32 connects to the Blynk IoT cloud platform through Wi-Fi. The measured temperature, humidity, soil moisture, light intensity, water level, and irrigation status can be monitored through a smartphone or web dashboard.

The IoT implementation provides remote access to greenhouse conditions and allows the user to observe the system without being physically present near the greenhouse.

9.5 AI Model Analysis

Historical sensor data collected from the greenhouse is used for training the Artificial Intelligence model. The model analyzes parameters such as temperature, humidity, soil moisture, and light intensity to identify patterns related to irrigation requirements.

The AI-based approach provides an additional decision-making layer over conventional threshold-based control. By using previously collected environmental data, the system can support more adaptive irrigation decisions according to the observed conditions.

9.6 Overall Result Analysis

The developed system demonstrates the integration of IoT and Artificial Intelligence for smart greenhouse automation. The prototype provides continuous environmental monitoring, automatic irrigation, remote Blynk monitoring, and data-based irrigation decision support.

The overall analysis shows that the system can reduce manual monitoring, avoid unnecessary irrigation, improve water management, and provide useful environmental information for greenhouse operation. The proposed system is therefore suitable for small-scale greenhouse applications, educational projects, home gardens, and further development in smart agriculture.

CHAPTER 10

ADVANTAGES AND APPLICATIONS

10.1 Introduction

The AI-Based Smart Greenhouse Automation System combines IoT, sensors, automatic irrigation, cloud monitoring, and Artificial Intelligence to provide efficient greenhouse management. The system offers several advantages in terms of water management, remote monitoring, reduced manual work, and intelligent irrigation control.

10.2 Advantages

The major advantages of the proposed system are:

- Automatic Irrigation: The water pump is automatically controlled based on soil moisture conditions.
- Reduced Water Wastage: Water is supplied only when the soil requires irrigation.
- Real-Time Monitoring: Temperature, humidity, soil moisture, light intensity, and water level can be monitored continuously.
- Remote Monitoring: Blynk IoT allows users to monitor greenhouse conditions through a smartphone or web dashboard.
- Reduced Manual Intervention: Automatic sensing and pump control reduce the need for continuous manual operation.
- AI-Based Decision Making: Historical sensor data is used to support intelligent and adaptive irrigation decisions.
- Water Level Monitoring: The system monitors the availability of water in the storage tank.
- Low-Cost System: The system uses commonly available and economical electronic components.
- Scalable Design: Additional sensors and control devices can be added according to application requirements.
- Improved Resource Utilization: The system helps in efficient utilization of water and electrical energy.

10.3 Applications

The proposed AI-Based Smart Greenhouse Automation System can be used in the following applications:

1. Home Gardens: Automatic irrigation and environmental monitoring for small home gardens.
2. Small-Scale Greenhouses: Monitoring and irrigation control for small agricultural greenhouses.
3. Educational Institutions: Demonstration of IoT, Artificial Intelligence, sensors, and automation concepts.
4. Research Greenhouses: Collection of environmental data for agricultural research and experimentation.
5. Nurseries: Monitoring soil moisture and environmental conditions for plant growth.
6. Smart Agriculture: Application of IoT and AI technologies for efficient agricultural practices.
7. Indoor Plant Cultivation: Automated monitoring and irrigation for controlled indoor growing environments.
8. Water Management Systems: Monitoring and controlling irrigation to improve efficient use of available water.

CHAPTER 11

APPENDICIES

CHAPTER 12

CONCLUSION

The AI-Based Smart Greenhouse Automation System was successfully designed and implemented to provide automatic greenhouse monitoring and irrigation. The system uses an ESP32 microcontroller along with DHT22, soil moisture, BH1750, and water level sensors to continuously monitor important environmental conditions.

The automatic irrigation system controls the water pump through a relay module based on the soil moisture condition, helping to reduce unnecessary water usage and manual intervention. The integration of the Blynk IoT platform enables users to monitor greenhouse conditions remotely through a smartphone or web dashboard.

The use of historical sensor data and an Artificial Intelligence model provides intelligent analysis of environmental conditions and supports adaptive irrigation decisions. Overall, the proposed system demonstrates how IoT and AI can be combined to improve greenhouse monitoring, irrigation management, and efficient resource utilization.

The system is cost-effective, scalable, and suitable for home gardens, educational projects, research greenhouses, nurseries, and small-scale agricultural applications. It provides a practical foundation for the development of more advanced smart agriculture systems in the future.

CHAPTER 13

REFERENCES

[1] Espressif Systems, *ESP32 Series Datasheet*, Version 5.3, 2026.

[2] Aosong Electronics Co., Ltd., *DHT22 (AM2302) Digital Temperature and Humidity Sensor Datasheet*, 2010.

[3] ROHM Semiconductor, *BH1750FVI Digital 16-bit Serial Output Type Ambient Light Sensor IC Datasheet*, 2011.

[4] Blynk Inc., *Blynk IoT Documentation and User Guide*, 2024.

[5] Arduino, *Arduino Programming Reference and Documentation*, 2024.

[6] Food and Agriculture Organization (FAO), *The State of Food and Agriculture: Leveraging Automation in Agriculture for Transforming Agrifood Systems*, 2022.

[7] J. G. Zambonelli et al., *Internet of Things (IoT) Applications in Smart Agriculture*, 2020.

[8] M. Ayaz, I. Ullah, Z. U. Khan, A. U. Rehman, and S. Kim, *IoT-Based Smart Agriculture: Applications, Challenges, and Future Directions*, 2019.

[9] A. Elijah, I. Rahman, A. Orikpete, T. A. Adebayo, and S. A. O. Akintayo, *A Comprehensive Review of Internet of Things (IoT) in Agriculture*, 2018.

[10] M. Shamshiri et al., *Advances in Greenhouse Automation and Intelligent Control Systems for Sustainable Agriculture*, 2018.